The Informational Cost of Spatial Prompts in SAM: A Controlled Study on Polyp Segmentation

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1. INTRODUCTION

Removing spatial prompts from SAM changes available localization and target specification. For polyp segmentation, diagnosis should distinguish missing mask candidates from failures to select a useful candidate. Building on SAM [1], prior polyp evaluations [2] and automatic prompting approaches [3, 4], we study the quality lost when spatial information is removed or degraded and assess learned replacements. We investigate informational loss and computational cost; human annotation time is not measured. Our contribution is controlled empirical evidence, rather than architectural novelty.

2. METHODOLOGY

We use 400 fixed training and 100 development images from Kvasir-SEG [5]. SAM ViT-B is frozen, with FP32 encoding, caching and decoding at longest-side input 1024. Conditions include empty spatial input, GT-derived interior points and boxes, controlled perturbations and decoder-output selection. Eligible GT components receive separate prompts and their masks are united, so oracle prompts can disclose component count. Automatic methods use no evaluation GT: four static tokens, an image-conditioned four-token generator, a linear segmentation probe over frozen SAM features, and probe-derived box/point prompts. Token initialization uses training GT through meanpoint. A fixed-prior control uses median normalized train-mask coordinates, with one box/point query per evaluated image.

Three additional RGB controls are trained from scratch: compact U-Net [6], a depthwise-separable U-shaped variant and FCN-Lite [7]. They are declared local variants, not exact paper reproductions. All initially use input 128, GroupNorm, BCE+Dice, Adam at 0.001, batch eight and flips for up to 30 epochs. A prespecified follow-up continues U-Net from identical epoch-30 weights, Adam and RNG states for 15 fixed epochs at 128 or 256. SAM token models use six epochs and a development-selected conditioned learning rate. Seeds 0/1/2 are used throughout; checkpoints and thresholds are selected on development data.

Dice is averaged per image at original resolution. Confidence intervals bootstrap images after averaging training seeds. A fixed-point-free permutation changes generator context while holding the target image, decoder and weights fixed. Latency uses decoded RGB, CPU/four threads, FP32, batch one, five warmups and twenty timed images; preprocessing, full network and restoration are included, while disk decoding and metrics are excluded. SAM timing includes its image encoder.

3. RESULTS AND DISCUSSION

Table 1 reports development scores. Probe-derived box/point prompts improve over the probe by 0.0690 Dice (paired 95% CI: 0.0471–0.0919). At the fixed 45-epoch endpoint, U-Net 256 reaches 0.8267 and SAM minus U-Net is −0.0088 [−0.0472, 0.0281]. The mean difference is unresolved by this interval, which also does not establish equivalence. Different pretraining and input sizes further limit causal architectural comparisons.

Table 1. Development results; GT prompts are oracle references.

Method	Development Dice [95% CI]
Empty spatial input, single output	0.0089 [0.0007, 0.0227]
GT point, oracle	0.6831 [0.6155, 0.7463]
GT box, oracle	0.9352 [0.9279, 0.9420]
Static tokens	0.3633 [0.2913, 0.4375]
Image-conditioned tokens	0.5013 [0.4260, 0.5762]
Linear probe	0.7489 [0.7067, 0.7879]
Probe → box+point → SAM	0.8179 [0.7707, 0.8600]
FCN-Lite, RGB 128	0.6002 [0.5549, 0.6429]
DS-U-Net, RGB 128	0.6548 [0.6122, 0.6946]
Compact U-Net, RGB 256, fixed epoch 45	0.8267 [0.7944, 0.8546]

Fixed train-derived box/point prompts reach 0.3582, versus 0.8179 for image-specific automatic localization (difference 0.4597 [0.3809, 0.5370]). Without prompts, even GT selection of the best of four decoder outputs reaches only 0.1824. With GT points, best-of-three reaches 0.8600 versus 0.5401 under predicted-IoU selection. Thus, candidate availability and selection limit different conditioning regimes.

Permuting generator context reduces Dice by only 0.0060 [0.0008, 0.0112], insufficient to attribute its full advantage over static tokens to matched image context. With 750 matched extra updates, the U-Net endpoint at 256 exceeds that at 128 by 0.0313 [0.0122, 0.0513]; this is adaptation from weights trained at 128, not two from-scratch recipes. Two 256 runs select the final epoch, so convergence remains unestablished. Median full-pipeline latency was 8.37 s for SAM automatic localization versus 27.08 ms for U-Net 256 on this CPU.

4. CONCLUSIONS

Image-specific automatic localization substantially improves frozen SAM over a fixed spatial prior. After strengthening the baseline, compact supervised segmentation is a competitive control whose mean quality difference from the SAM pipeline remains uncertain. These findings characterize conditioning dependence without isolating an intrinsic decoder defect. Development guided design and selection, the historical test was already inspected, and no new independent evaluation was performed. Label-budget and external evaluations are needed before stronger claims.

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